Detecting AI-Generated Product Images Using Deep Learning

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Abstract — The rapid advancement of generative artificial intelligence has made it increasingly difficult to distinguish between authentic product images and synthetically generated content. This poses significant challenges for online marketplaces such as Etsy, where trust in product authenticity is critical. In this study, we investigate the effectiveness of deep learning models for binary image classification, aiming to detect AI-generated images. Multiple pre-trained convolutional neural network architectures, including ResNet50 and DenseNet121, are evaluated using a transfer learning approach. A two-stage training strategy, combined with data augmentation and threshold optimization, is employed to improve model performance. The proposed approach achieves a peak F1-score of 0.83 on the validation set using ResNet50, demonstrating the effectiveness of fine-tuned deep learning models for AI image detection tasks.

Keywords —*AI Image Detection, Deep Learning, Transfer Learning, Convolutional Neural Networks, Image Classification, ResNet50, DenseNet121, Threshold Optimization, Computer Vision, Digital Image Forensics*

I. INTRODUCTION

The rapid advancement of generative artificial intelligence has significantly transformed the creation and distribution of digital media. Modern generative models, including Generative Adversarial Networks (GANs) and diffusion-based architectures, are capable of producing highly realistic images that closely resemble authentic photographs. While these technologies enable innovative applications in design and content creation, they also introduce serious challenges related to digital authenticity and trust.

In online marketplaces such as Etsy, product listings rely heavily on visual content to inform purchasing decisions. However, the use of AI-generated images in place of real product photographs can mislead consumers, creating discrepancies between advertised and actual products. As the visual quality of synthetic images continues to improve, manual verification becomes increasingly unreliable and inefficient. This highlights the need for automated systems capable of distinguishing between authentic and AI-generated images.

This problem can be formulated as a binary image classification task, where each image is classified as either authentic or AI-generated. Deep learning, particularly convolutional neural networks (CNNs), has demonstrated strong performance in image classification tasks due to its ability to learn hierarchical feature representations. Transfer learning further enhances this capability by leveraging models pre-trained on large-scale datasets and adapting them to domain-specific tasks.

In this study, we investigate the effectiveness of multiple pre-trained CNN architectures for detecting AI-generated images. Specifically, we explore models such as ResNet50 and DenseNet121, applying a two-stage transfer learning strategy involving feature extraction and fine-tuning. Additionally, we incorporate data augmentation and decision threshold optimization to improve model generalization and classification performance. The objective is to evaluate how different architectures and training strategies impact the detection of subtle artifacts present in AI-generated images.

II. RELATED WORK

A. Detection Methodologies for synthetic Media

The detection of AI-generated content has evolved from traditional digital forensics to deep learning-based feature extraction. Martin-Rodriguez et al. [2] highlight that while natural images contain unique camera sensor "fingerprints" such as Photo Response Non-Uniformity (PRNU), synthetic images often lack these, instead exhibiting specific pixel-level artifacts that can be identified using Error Level Analysis (ELA) and CNNs. This is further supported by Lokner Lađević et al. [1], who demonstrate that synthetic and authentic images exhibit fundamentally different spatial activations. Using Explainable AI (XAI) tools like Grad-CAM++, they found that CNNs typically focus on the main object in real photographs, whereas in AI-generated images, the model's "attention" is often diverted to backgrounds and complex textures where generative artifacts are most prevalent [1].

B. The Efficacy of transfer Learning and Fine-Tuning

Transfer learning has become the standard for binary image classification when domain-specific data is limited. Dewan et al. [4] and Mahmoud et al. [3] establish that pre-trained architectures such as ResNet50, DenseNet, and EfficientNet offer a significant advantage by leveraging hierarchical features learned from large-scale datasets like ImageNet. Their comparative studies show that while architectures like DenseNet excel at feature reuse in low-data environments [4], the process of fine-tuning is what truly adapts these models to specialized tasks. Specifically, the two-stage strategy of "warm-up" training (freezing the base) followed by unfrozen fine-tuning allows the network to preserve general spatial hierarchies while learning the subtle, low-level "textures" unique to generative AI [3].

C. Robust Evaluation and decision Threshold

Calibration A critical consensus across current literature is that standard accuracy is an insufficient metric for high-stakes binary classification, such as detecting fraudulent product listings. Sujon et al. [5] utilize advanced statistical validation to argue that metrics like the F1-score and Matthews Correlation Coefficient (MCC) provide a more stable assessment of model reliability, especially in

datasets where classes may be imbalanced. This aligns with the work of Lokner Lađević et al. [1], who emphasize the need for transparency via XAI to ensure the model isn't overfitting to noise. Furthermore, Sujon et al. [5] demonstrate that "threshold sensitivity analysis"—adjusting the decision boundary away from the default 0.5—is essential for optimizing the trade-off between precision and recall, a technique that directly validates the threshold optimization methodology used in this study.

III. METHODOLOGY

A. Dataset and Data Preparation

The experimental framework utilizes a dataset comprising 4,800 labeled training images and 2,058 unlabeled test images. As real-world datasets often contain inconsistencies between metadata and physical files, a verification step was incorporated into the pipeline. A validation script was used to ensure that each image referenced in the CSV file corresponded to a valid file within the dataset directory.

Any missing or inaccessible files were identified and excluded to maintain dataset integrity. The verified dataset was then partitioned using an 80/20 stratified split, resulting in 3,840 training samples and 960 validation samples. Stratification ensured that the class distribution between authentic and AI-generated images remained consistent across both subsets, providing a reliable basis for model evaluation.

B. Augmentation and Normalization

To improve generalization performance and reduce overfitting, a comprehensive data augmentation strategy was applied using the Keras ImageDataGenerator API. This approach introduces controlled variations in the training data, encouraging the model to learn robust and meaningful features rather than memorizing specific visual patterns.

The augmentation pipeline includes both geometric and photometric transformations applied dynamically during training. Geometric transformations consist of random rotations, horizontal flipping, and spatial shifts, which help the model become invariant to positional and orientation changes. Photometric transformations include brightness adjustments and zoom operations, enabling the model to handle variations in lighting conditions and scale.

All input images were resized to a fixed resolution of 224 × 224 × 3 to ensure compatibility with pre-trained network architectures. Additionally, pixel values were normalized using model-specific preprocessing functions, aligning the input distribution with the expectations of the corresponding backbone networks. For validation and test datasets, only resizing and normalization were applied to maintain consistent and unbiased evaluation conditions.

C. Architectural Exploration

Recognizing that different network topologies capture distinct hierarchical features, the study conducts a comparative analysis across several deep learning architectures. The backbones act as feature extractors, leveraging weights pre-trained on the ImageNet dataset:

- ResNet50 & DenseNet121: Explored as strong traditional baselines, leveraging residual connections and dense feature concatenation to detect low-level synthetic anomalies.
- EfficientNet (B0 and B4): Utilized for their compound scaling rules, providing a highly optimized balance between network depth, width, and resolution.
- ConvNeXt (Tiny): Explored as a modern, modernized standard CNN that competes directly with Vision Transformers, offering superior transfer learning capabilities for subtle texture and artifact recognition.

Custom Classification Head: Regardless of the backbone, the top fully connected layers of the original models are removed. A custom classification head is appended to tailor the network for binary detection. This head typically consists of a GlobalAveragePooling2D layer to reduce spatial dimensions, followed by BatchNormalization to stabilize learning, and a Dropout layer (varying between 0.2 and 0.5) to introduce heavy regularization. The network terminates in a single Dense node utilizing a Sigmoid activation function to output a continuous probability score between 0.0 and 1.0.

D. Multi-Stage Training Strategy

To safely adapt the ImageNet-trained weights to the specific domain of AI image detection, a two-phase transfer learning methodology is employed:

- **Phase 1: Feature Extraction (Warm-up):** The convolutional base is entirely frozen (trainable = False), protecting the pre-trained spatial hierarchies from being destroyed by large random gradients. The model is compiled using the Adam optimizer with a relatively high learning rate (le10^{-3} or 10^{-4}). The custom classification head is trained for an initial set of epochs (typically 5 to 8) to map the extracted features to the binary labels.
- **Phase 2: Fine-Tuning:** The network is partially or fully unfrozen (e.g., unfreezing the top 30-50 layers or the entire base model). The learning rate is drastically reduced by an order of magnitude (10^{-5}) to allow for subtle weight updates. This allows the model to learn the highly specific, low-level "textures" and blending errors unique to AI generators without catastrophic

E. Optimization and Evaluation Metrics

Throughout both training phases, EarlyStopping was used to monitor validation loss, halting training when no improvement was observed over a predefined patience period. ModelCheckpoint was employed to ensure that only the model weights corresponding to the best validation performance were retained.

Model evaluation extended beyond standard accuracy metrics. Performance was assessed using Precision, Recall, and F1-score, providing a more comprehensive understanding of the model's ability to balance false positives and false negatives.

To further improve performance, threshold optimization was applied. Instead of using a fixed classification threshold of 0.5, an iterative search was conducted over a range of threshold values from 0.3 to 0.7, with increments of 0.02. The threshold that maximized the F1-score on the validation set was selected as the optimal decision boundary. This approach enabled better calibration of model predictions prior to generating final outputs on the test dataset.

IV. RESULTS

A. Baseline Model Performance

An initial baseline was established using a Random Forest classifier. To construct the feature space, images were downsampled to 32 × 32 pixels and flattened into one-dimensional vectors. This traditional machine learning approach achieved an overall accuracy of 66%, with a precision of 0.6480, recall of 0.6242, and an F1-score of 0.6359.

While the model demonstrated performance above random chance, it struggled to capture the complex and high-dimensional features present in AI-generated images. The results highlight the limitations of traditional methods and emphasize the need for deep learning approaches capable of extracting richer spatial and textural representations.

B. Fine-Tuning Performance

Subsequent experiments utilized pre-trained convolutional neural network architectures initialized with ImageNet weights. In this phase, the convolutional base was frozen, and only the custom classification head was trained using the Adam optimizer with a learning rate in the range of 1e-4 to 1e-3.

EfficientNetB0 achieved a validation accuracy of approximately 77%, with an F1-score of 0.7619, demonstrating strong baseline performance among the evaluated models. ConvNeXt Tiny also performed competitively, achieving a validation accuracy of 76.15% after five epochs.

In comparison, ResNet50 and DenseNet121 achieved validation accuracies of 71.56% and 70.63%, respectively, indicating limited effectiveness when relying solely on pre-trained feature representations. EfficientNetB4 exhibited the weakest performance in this phase, achieving only 59.06% accuracy, suggesting challenges in training larger architectures under constrained conditions.

These results indicate that while pre-trained models provide strong general feature extraction capabilities, feature extraction alone is insufficient for capturing the subtle artifacts present in AI-generated images, highlighting the importance of fine-tuning.

As illustrated in Fig. 1, the overall pipeline consists of preprocessing, feature extraction using a pre-trained CNN, and a custom classification head.

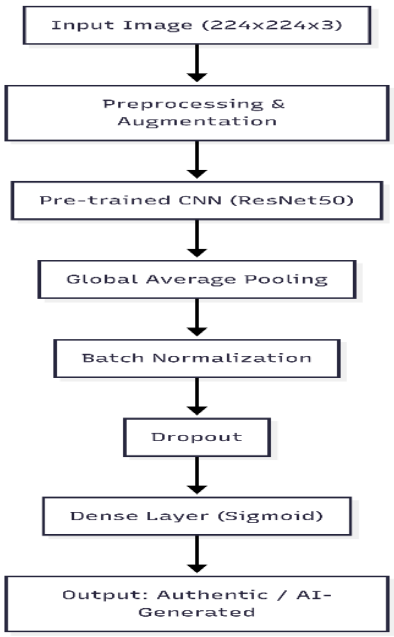


Fig. 1. Overview of the proposed ResNet50-based deep learning pipeline for AI-generated image detection.

The pipeline illustrates the end-to-end workflow, beginning with image preprocessing and augmentation, followed by feature extraction using a pre-trained ResNet50 backbone. The extracted features are then passed through a custom classification head consisting of global average pooling, batch normalization, and dropout, before producing a final binary prediction via a sigmoid activation function.

C. Threshold Optimization

To further improve classification performance, threshold optimization was applied to refine the decision boundary of the models. Instead of relying on a fixed classification threshold of 0.5, a range of threshold values between 0.3 and 0.7 was evaluated in increments of 0.02 to identify the value that maximized the F1-score on the validation set.

For the ResNet50 model, the optimal threshold was found to be 0.68. Adjusting the threshold from the default value of 0.5 significantly reduced false positives, resulting in an improvement in F1-score from approximately 0.79 to 0.83. This indicates that the model initially exhibited a bias toward predicting AI-generated images, leading to higher recall but lower precision.

By increasing the decision threshold, the model became more conservative in predicting the AI-generated class, improving the balance between precision and recall. This adjustment enhanced overall classification reliability, particularly by reducing incorrect predictions of authentic images.

A similar threshold tuning process was applied to other models; however, the most significant improvement was

observed in ResNet50. These results demonstrate that post-processing techniques such as threshold calibration can substantially enhance model performance without requiring modifications to the underlying architecture.

The classification performance can be further analyzed using the confusion matrix, as shown in Fig. 2.

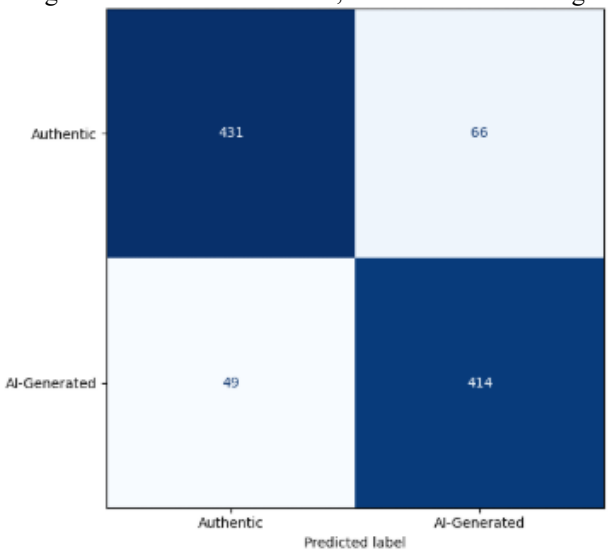


Fig. 2. Confusion Matrix for the ResNet50 model on the validation set.

The confusion matrix demonstrates that the model achieves high recall for AI-generated images, correctly identifying the majority of synthetic samples. However, it also reveals a tendency to misclassify certain authentic images as AI-generated, indicating a bias toward the positive class. This behavior reflects a trade-off between precision and recall, where the model prioritizes detecting AI-generated content at the cost of increased false positives.

D. Comparative Analysis

A comparative evaluation of all models reveals clear differences in performance based on architecture and training strategy. Among the evaluated approaches, ConvNeXt Tiny achieved the highest overall performance, with an F1-score of 0.8780 and an accuracy of 88%, demonstrating its effectiveness in capturing fine-grained image features. EfficientNetB0 also performed strongly, achieving an F1-score of 0.8636, indicating that compound scaling contributes to robust feature extraction.

ResNet50, while achieving a lower overall F1-score of 0.83, demonstrated significant improvement after threshold optimization. Its initially high recall (0.94) and lower precision highlighted a bias toward predicting AI-generated images, which was effectively mitigated through decision boundary adjustment. This suggests that while the architecture was capable, its performance was highly sensitive to post-processing techniques.

DenseNet121 achieved comparatively lower performance, with an F1-score of 0.75, indicating that dense feature reuse alone was insufficient for detecting the subtle inconsistencies present in AI-generated images. Similarly, EfficientNetB4 underperformed, likely due to its increased complexity and difficulty in converging effectively within the given training constraints.

Overall, the results indicate that modern architectures such as ConvNeXt and EfficientNet outperform traditional convolutional networks in this task. However, the findings also highlight that training strategy and threshold optimization play a critical role in achieving optimal performance, often providing substantial gains without changes to the underlying model architecture.

Model	Threshold	Precision	F1-score	Recall	Accuracy
Random Forest (Baseline)	0.50	0.65	0.64	0.62	66%
EfficientNetB4	0.32	0.73	0.65	–	62%
DenseNet121	0.40	0.71	0.76	0.77	75%
ResNet50	0.68	0.71	0.83	0.94	79%
EfficientNetB0	0.50	0.87	0.86	0.86	87%
ConvNeXt Tiny	0.50	0.86	0.88	0.89	88%

TABLE I. Comparative Model Performance on Validation Set

Table I summarizes the performance of all evaluated models across key metrics, highlighting the impact of architecture choice and optimization techniques on classification performance..

V. CONCLUSIONS AND FUTURE WORK

This study explored the effectiveness of deep learning models for detecting AI-generated product images in an e-commerce setting. Multiple architectures, including ResNet50, DenseNet121, EfficientNet, and ConvNeXt, were evaluated using a transfer learning approach combined with data augmentation and threshold optimization.

Among the evaluated models, ConvNeXt Tiny achieved the highest performance, with an F1-score of 0.8780 and an accuracy of 88%, followed closely by EfficientNetB0. ResNet50 also demonstrated strong performance, achieving an F1-score of 0.83 after threshold optimization, highlighting the importance of post-processing techniques in improving model predictions.

The results indicate that while modern architectures provide strong feature extraction capabilities, model performance is significantly influenced by training strategy and decision threshold calibration. In particular, threshold optimization proved effective in reducing false positives and improving the balance between precision and recall.

Future work may involve exploring ensemble methods to combine the strengths of multiple architectures, as well as incorporating explainable AI techniques to better understand model decisions. Additionally, evaluating performance on more diverse datasets and newer generative models would further enhance the robustness of the approach.

VI. REFERENCES

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